

Technology #01: Multi-Segment Flow Control with Bayesian Occlusion Prediction

Technology Domain: Hydraulics + Control Systems

Mechanism: Multi-segment CSF shunt catheter with per-segment flow sensors, Bayesian occlusion predictor, and alpha-distribution controller

Maturity: ENGINEERING_DEFINITION

Current State: No physical prototype. Computational model and engineering definition exist.

Key Risk: Multi-segment shows no advantage over single-segment in $\geq 80\%$ of computational scenarios

Next Experiment: V0 bench prototype (4-segment catheter + COTS sensors + Arduino)

Buyer Receives: Multi-segment catheter concept + mechanism description; Bayesian predictor algorithm (225 ablation runs, rate-limiting confirmed); Alpha-distribution controller with graceful degradation protocol; Dual-invariant falsification record (honest failure disclosure); V0 bench prototype BOM (COTS sensors + Arduino); Computational model (svMultiPhysics verified 1D within 16%); This engineering dossier with external precedent

Buyer Must Build: Production catheter design (dimensioned drawings, tolerances); Implantable-grade flow sensor (COTS $\rightarrow$ implantable transition); Production controller (Arduino $\rightarrow$ medical-grade MCU); Manufacturing process for multi-segment catheter; Regulatory submission (PMA Class III); Clinical validation evidence

Transfer Posture: Sponsored Validation

Target Buyer: Neuroshunt manufacturer (Medtronic, Integra, Miethke)

CONFIDENTIAL. See full engineering dossier for details.